

ActiveSIGMA: Semantic Inconsistency-Guided Active Mapping

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Abstract—Autonomous robots operating in unknown environments require exploration to construct maps with accurate geometry and reliable semantic understanding for downstream tasks such as navigation, object search, and human-robot interaction. Existing active semantic mapping methods mainly select viewpoints according to geometric coverage, photometric reconstruction quality, or semantic uncertainty, but do not explicitly address cross-view semantic inconsistency, where the same 3D region receives conflicting semantic predictions from different viewing directions. To address this problem, we propose *ActiveSIGMA*, a semantic inconsistency-guided active mapping framework based on a hybrid Gaussian-voxel representation. The Gaussian map supports dense reconstruction and rendering, while the voxel map stores direction-conditioned semantic evidence. *ActiveSIGMA* converts cross-view semantic inconsistency into directional exploration heat to prioritize semantically inconsistent and insufficiently observed regions during candidate generation and next-best-view selection. Experiments on Replica and Matterport3D (MP3D) datasets demonstrate that *ActiveSIGMA* improves semantic mapping accuracy over existing methods on both datasets and produces more consistent semantic predictions across novel viewpoints.

I. INTRODUCTION

Active mapping enables autonomous robots to select informative viewpoints while constructing maps of unknown environments [1], [2]. This capability is important for downstream tasks such as navigation, object search, and human-robot interaction, where the map should provide both accurate geometry and reliable semantic understanding. Recent advances in 3D Gaussian Splatting (3DGS) enable efficient dense scene reconstruction and novel-view rendering [3], and have been adopted for online dense SLAM and mapping systems [4], [5]. This rendering capability also makes 3DGS well suited to active mapping, because candidate viewpoints can be evaluated by rendering the current map from hypothetical camera poses. Accordingly, recent 3DGS-based active mapping methods select the next-best-view (NBV) using geometric incompleteness, rendering uncertainty, photometric quality, visibility, and information gain [6], [7], [8].

More recently, active mapping has been extended beyond geometric and photometric reconstruction by incorporating semantic information into viewpoint selection. For example, ActiveSGM [9] uses semantic uncertainty to estimate the informativeness of candidate observations and prioritize regions whose semantic predictions remain ambiguous. However, semantic uncertainty primarily estimates ambiguity within individual semantic predictions and do not explicitly capture how predictions for the same 3D region vary across viewing directions.

Semantic predictions can vary across viewpoints because of occlusion, partial observation, and view-dependent appearance, causing the same 3D region to receive inconsistent class distributions from different viewing directions [10], [11]. Multi-view fusion can improve semantic mapping quality [12], while viewpoint consistency has also been studied for active semantic segmentation [13]. Nevertheless, existing 3DGS-based active semantic mapping methods evaluate semantic informativeness primarily from individual candidate views, leaving cross-view variations underexplored when deciding where and from which direction to acquire additional observations. Consequently, per-view semantic uncertainty alone does not reveal such directional conflicts or indicate the corresponding direction that requires further observation.

To address this limitation, we propose *ActiveSIGMA*, a semantic inconsistency-guided active mapping framework for 3DGS. *ActiveSIGMA* maintains a hybrid Gaussian-voxel map, where the Gaussian map supports dense scene reconstruction and rendering, while the voxel map stores direction-conditioned Dirichlet semantic evidence [14], [15]. For each voxel and discretized direction, the Dirichlet evidence encodes the semantic distribution and accumulated observation strength, enabling the system to distinguish between under-observed regions and regions with cross-view semantic inconsistency. We convert this inconsistency into directional exploration heat, which guides semantic refinement target selection, candidate generation, and NBV scoring. Our contributions are:

- We propose *ActiveSIGMA*, an active semantic mapping framework that explicitly identifies and revisits regions with cross-view semantic inconsistency during exploration.
- We introduce a hybrid Gaussian-voxel representation that preserves direction-conditioned semantic distributions and accumulated observation evidence across viewpoints.
- We formulate directional exploration heat for candidate generation and next-best-view (NBV) scoring to prioritize semantically inconsistent and under-observed voxel-direction pairs.
- Experimental results on Replica and MP3D show that *ActiveSIGMA* improves semantic mapping over existing methods, maintains comparable geometric and photometric reconstruction quality, and produces more consistent semantic predictions across novel viewpoints.

* Denotes equal contribution.

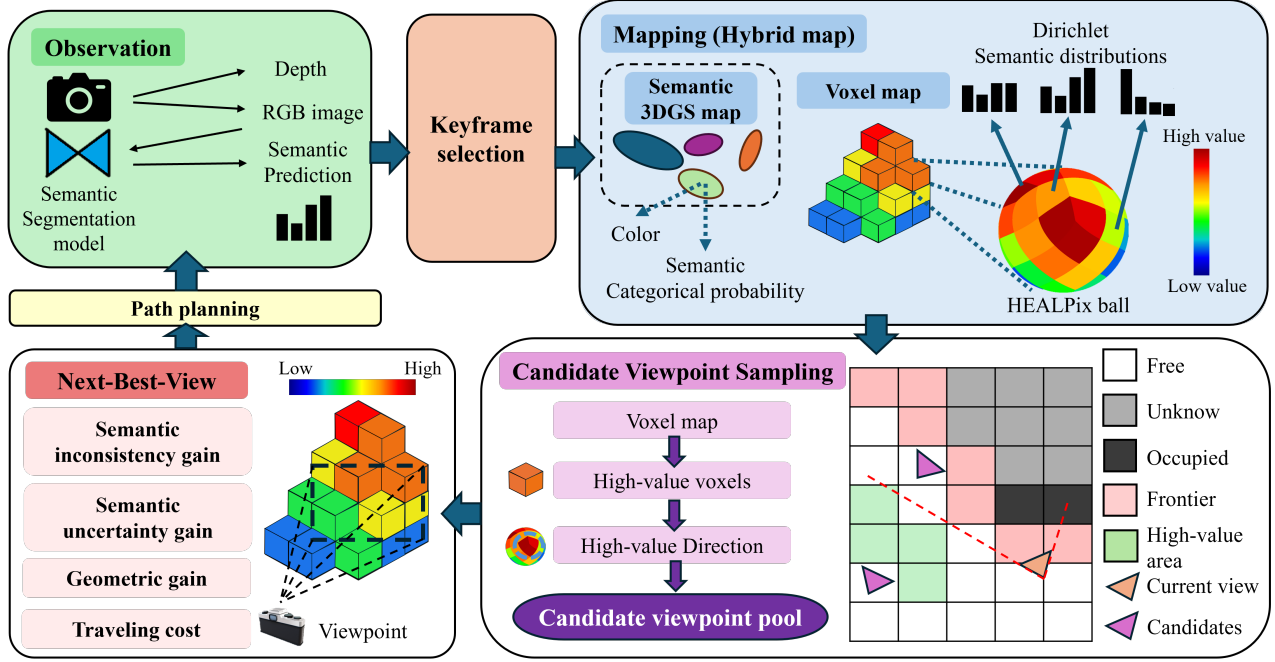


Fig. 1. Overview of the proposed ActiveSIGMA pipeline. Posed RGB-D observations and semantic predictions update a hybrid map consisting of a semantic 3D Gaussian map for dense reconstruction and an exploration voxel map for storing direction-conditioned semantic evidence. The accumulated evidence is converted into exploration values for semantic refinement candidate generation and next-best-view selection.

II. METHOD

A. Overview

We propose ActiveSIGMA, a semantic inconsistency-guided active mapping framework that jointly maintains a semantic 3D Gaussian map and an exploration-oriented voxel map. Posed RGB-D observations and OneFormer [16] semantic predictions update both maps. The Gaussian map supports dense scene reconstruction and candidate-view rendering, while the voxel map stores occupancy and direction-conditioned semantic evidence. The planner uses this evidence to compute voxel-level exploration values, generate geometric candidates from reachable free space and semantic refinement candidates from high-value voxel-direction pairs, and select the NBV by balancing semantic inconsistency gain, semantic uncertainty gain, geometric exploration gain, and travel cost. The overall pipeline is illustrated in Fig. 1.

B. Hybrid Gaussian-Voxel Map

To support both dense scene rendering and structured exploration reasoning, we maintain a hybrid map $(\mathcal{G}, \mathcal{M})$. The semantic Gaussian map $\mathcal{G}$ represents geometry, appearance, opacity, and semantic attributes for efficient RGB-D and semantic rendering, whereas the voxel map $\mathcal{M}$ provides a discrete planning representation that stores semantic evidence conditioned by occupancy and direction $\alpha_{v,d}$ in HEALPix [17] direction bins d . This hybrid map representation supports candidate evaluation through Gaussian rendering, while semantic inconsistency and exploration value are computed from structured evidence stored in the voxel map.

C. Direction-Conditioned Semantic Evidence

To capture cross-view semantic inconsistency, we maintain semantic evidence across observation directions. Given a posed RGB-D frame at time t , OneFormer predicts a semantic probability vector $\pi_t(u) \in \mathbb{R}^{N_c}$ for each pixel u . Valid depth pixels are back-projected into the voxel map. Let $\Omega_t(v)$ denote the set of pixels assigned to voxel v . Their semantic predictions are aggregated into a voxel-level observation:

$$\mathbf{o}_t(v) = \frac{\sum_{u \in \Omega_t(v)} \omega_t(u) \pi_t(u)}{\sum_{u \in \Omega_t(v)} \omega_t(u) + \epsilon}, \quad (1)$$

where $\omega_t(u)$ denotes the pixel-level reliability weight calculated from the confidence of the observed semantic prediction and its agreement with the rendered semantics to reduce the influence of unreliable predictions, and ϵ is a small constant. The mean pixel-level reliability within voxel v is $\bar{\omega}_t(v)$ and the corresponding voxel-level reliability is defined as $\rho_t(v) = 0.3 + 0.7 \cdot \text{clip}(\bar{\omega}_t(v), 0, 1)$.

Each voxel observation is assigned to a HEALPix direction bin according to its camera-to-voxel viewing direction. Let $\mathbf{t}_t$ and $\mathbf{c}_v$ denote the camera and voxel centers, respectively. The assigned direction bin is

$$d_t(v) = b \left(\frac{\mathbf{c}_v - \mathbf{t}_t}{\|\mathbf{c}_v - \mathbf{t}_t\|_2 + \epsilon} \right), \quad (2)$$

with $b(\cdot)$ mapping the viewing direction to the corresponding HEALPix bin.

To represent the directional semantic distribution and the accumulated evidence, we model the semantic prediction for each voxel-direction pair using a Dirichlet distribution. For each voxel v and direction bin d , the voxel map stores a

Dirichlet parameter vector $\alpha_{v,d} \in \mathbb{R}^{K+1}$, initialized at $t = 0$ with a small symmetric prior $\alpha_{v,d}^0$. All bins of a voxel share a fixed top- K semantic class basis determined by the first valid observation of that voxel. Subsequent observations are projected onto this basis, with remaining probability mass assigned to a residual component. The resulting observation is denoted by $\tilde{\mathbf{o}}_t(v) \in \mathbb{R}^{K+1}$ and the evidence associated with the observed direction is updated as

$$\alpha_{v,d_t(v)}^t = \alpha_{v,d_t(v)}^{t-1} + \rho_t(v) \tilde{\mathbf{o}}_t(v), \quad (3)$$

The accumulated evidence strength and the corresponding directional semantic distribution are

$$\mu_{v,d} = \frac{\alpha_{v,d}}{S_{v,d}}, \quad S_{v,d} = \sum_{\ell=1}^{K+1} \alpha_{v,d,\ell}. \quad (4)$$

Here, $S_{v,d}$ quantifies the amount of semantic evidence accumulated from direction d , while $\mu_{v,d}$ represents the corresponding semantic distribution. Maintaining these quantities separately across direction bins preserves view-dependent semantic observations and enables cross-view inconsistency to be measured.

D. Directional Exploration Heat

Given direction-conditioned evidence, we assign exploration heat to each voxel-direction pair to identify directions that are semantically inconsistent and insufficiently observed.

For voxel v and direction bin d , we measure the discrepancy between its semantic distribution and those of the adjacent HEALPix bins using Jensen-Shannon divergence [18]:

$$D_{v,d} = \frac{1}{|\mathcal{N}(d)|} \sum_{d' \in \mathcal{N}(d)} \text{JSD}(\mu_{v,d}, \mu_{v,d'}), \quad (5)$$

$$H_{v,d} = m_{v,d} D_{v,d} \frac{1}{S_{v,d} + 1}.$$

where $\mathcal{N}(d)$ denotes adjacent HEALPix bins. The inconsistency score $D_{v,d}$ is high when the semantic prediction at d disagrees with nearby observations, while $1/(S_{v,d} + 1)$ prioritizes directions with limited accumulated evidence. The feasibility mask $m_{v,d} \in [0, 1]$ suppresses voxel-direction pairs without sufficient free space for candidate placement.

Let $\mathcal{D}_v$ denote the observed direction bins with at least one valid semantic observation. The mean exploration heat $\bar{H}_v$, normalized semantic entropy $\hat{E}_v$, and voxel value $V(v)$ are defined as

$$\bar{H}_v = \frac{1}{|\mathcal{D}_v|} \sum_{d \in \mathcal{D}_v} H_{v,d},$$

$$\hat{E}_v = \frac{1}{|\mathcal{D}_v|} \sum_{d \in \mathcal{D}_v} \frac{-\sum_{\ell=1}^{K+1} \mu_{v,d,\ell} \log \mu_{v,d,\ell}}{\log(K+1)}, \quad (6)$$

$$V(v) = \lambda_h \bar{H}_v + \lambda_u \hat{E}_v.$$

Here, $\bar{H}_v$ captures cross-view inconsistency, while $\hat{E}_v$ measures semantic uncertainty. Their contributions are balanced by λ_h and λ_u . Voxels with high $V(v)$ and their high-heat directions are used as semantic refinement targets.

E. Candidate Viewpoint Generation and NBV Selection

The planner constructs a candidate pool $\mathcal{C}$ by combining geometric exploration candidates sampled from newly discovered free space and semantic refinement candidates generated from high-value voxel-direction pairs.

For semantic refinement candidates, we select the top- K_v voxels according to $V(v)$ and retain their top- K_d direction bins according to the exploration heat $H_{v,d}$. Let $\mathbf{q}_d$ denote the unit camera-to-voxel direction represented by bin d and r_i denotes the sampled camera-to-voxel distance. Candidate camera centers are sampled as $\mathbf{t}_i = \mathbf{c}_v - r_i \mathbf{q}_d$ and oriented toward the corresponding voxel center. Candidates that are not in free space or cannot be reached by the planner are discarded.

For each candidate $T_i \in \mathcal{C}$, we render the Gaussian map and back-project the rendered depth map into the voxel map to obtain the set of visible voxels $\mathcal{V}(T_i)$. The candidate is then evaluated using semantic inconsistency gain $G_I(T_i)$, semantic uncertainty gain $G_U(T_i)$, geometric exploration gain $G_g(T_i)$, and travel cost $C(T_i)$.

Semantic inconsistency gain. This term measures the expected benefit of observing voxels from directions with high exploration heat:

$$G_I(T_i) = \frac{\sum_{v \in \mathcal{V}(T_i)} w_v H_{v,d_{T_i}(v)}}{\sum_{v \in \mathcal{V}(T_i)} w_v + \epsilon}, \quad (7)$$

where $w_v \in [0, 1]$ is obtained by aggregating the rendered opacity of the visible Gaussians associated with voxel v , and $d_{T_i}(v)$ is the HEALPix direction bin from which candidate T_i observes the voxel v .

Semantic uncertainty gain. This term encourages observing voxels whose semantic distributions remain uncertain using the normalized directional entropy $\hat{E}_v$ defined in (6):

$$G_U(T_i) = \frac{\sum_{v \in \mathcal{V}(T_i)} w_v \hat{E}_v}{\sum_{v \in \mathcal{V}(T_i)} w_v + \epsilon}. \quad (8)$$

Geometric exploration gain. Following ActiveSGM, we measure the geometric exploration gain by counting missing pixels in the rendered silhouette:

$$G_g(T_i) = \sum_{u \in \mathcal{U}} \mathbf{1}[M_{T_i}(u) = 0]. \quad (9)$$

where $\mathcal{U}$ denotes the candidate-view image domain and $M_{T_i}(u) = 1$ indicates valid rendered support at pixel u .

Travel cost. This term is approximated by the Euclidean distance from the current position $\mathbf{t}_{cur}$ to the candidate camera center $\mathbf{t}_{T_i}$ as $C(T_i) = \|\mathbf{t}_{T_i} - \mathbf{t}_{cur}\|_2$.

NBV scoring. All gains and travel costs are min-max normalized over the candidate pool $\mathcal{C}$. Let $\tilde{G}_{g,i}$, $\tilde{G}_{I,i}$, $\tilde{G}_{U,i}$, and $\tilde{C}_i$ be normalized terms. The final NBV score is

$$S_i = \lambda_g \tilde{G}_{g,i} + \lambda_I \tilde{G}_{I,i} + \lambda_U \tilde{G}_{U,i} - \lambda_C \tilde{C}_i. \quad (10)$$

where $\lambda_g, \lambda_I, \lambda_U, \lambda_C \geq 0$ control the relative contribution of each term. The selected NBV,

$$T^* = \arg \max_{T_i \in \mathcal{C}} S_i,$$

is executed using an RRT-based path planner [19].

TABLE I
QUANTITATIVE COMPARISONS ON REPLICA AND MP3D.

Dataset	Method	Avg. Steps	PSNR $\uparrow$	RMSE $\downarrow$	mIoU $\uparrow$	mAP $\uparrow$
Replica	SGS-SLAM	2000	27.13	0.0708	80.45	89.98
Replica	ActiveSGM †	783	30.46	0.0138	84.48	93.22
Replica	Ours	768	30.83	0.0124	84.86	94.33
MP3D	ActiveSGM †	2325	26.93	0.0190	74.54	86.72
MP3D	Ours	2007	26.80	0.0175	74.69	86.88

† Reproduced using the official implementation and evaluated at the end of active exploration, consistent with the evaluation stage used for ActiveSIGMA.

TABLE II
ABLATION STUDY OF NBV SCORING COMPONENTS.

Ablation	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	RMSE $\downarrow$	mIoU $\uparrow$	mAP $\uparrow$
w/o geom. gain	30.26	0.9522	0.1553	0.0164	84.35	93.05
w/o sem. incons. gain	30.61	0.9565	0.1520	0.0120	84.10	94.02
w/o sem. uncert. gain	30.70	0.9572	0.1455	0.0139	84.58	94.24
w/o semantic gains	30.01	0.9525	0.1530	0.0114	83.94	93.14
Ours	30.83	0.9579	0.1465	0.0124	84.86	94.33

III. EXPERIMENTS

A. Experimental Setup

We evaluate ActiveSIGMA on eight Replica [20] scenes and four Matterport3D (MP3D) [21] scenes. SGS-SLAM [22] is included as a passive semantic mapping reference on Replica, while ActiveSGM [9] is the primary active mapping baseline on both datasets. We reproduce ActiveSGM using its official implementation and compare the maps produced at the end of active exploration with ActiveSIGMA. ActiveSGM and ActiveSIGMA use the same semantic 3DGS backend, OneFormer semantic predictions, comparable exploration steps, RRT-based planner, and held-out evaluation views. All reconstructed maps are evaluated

by PSNR (dB) for photometric quality, depth RMSE (m) for geometric accuracy, and mIoU (%) and mAP (%) for semantic mapping performance.

B. Quantitative Results

Table I reports the quantitative results. On Replica, ActiveSIGMA achieves higher average PSNR, mIoU, and mAP and lower depth RMSE than ActiveSGM. On MP3D, ActiveSIGMA improves depth RMSE, mIoU and mAP, while ActiveSGM obtains a higher PSNR. Overall, ActiveSIGMA improves semantic mapping accuracy on both datasets while maintaining comparable geometric and photometric reconstruction quality.

C. Qualitative Comparison

Fig. 2 and Fig. 3 compare novel-view semantic renderings on Replica and MP3D. ActiveSIGMA produces more consistent semantic predictions across views, whereas ActiveSGM changes the predicted class under viewpoint variation. These results show that direction-conditioned semantic evidence helps identify and reduce cross-view semantic inconsistency.

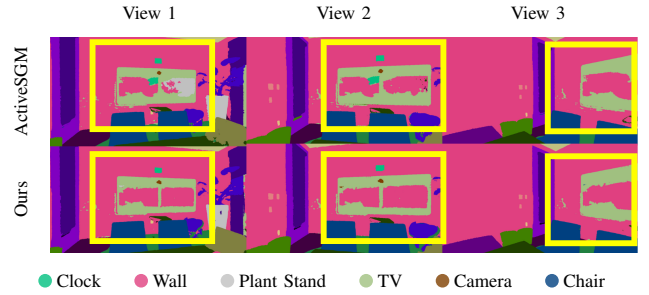


Fig. 2. Novel-view semantic rendering comparison in Replica. The yellow boxes highlight a screen region where ActiveSGM changes labels across views, while ActiveSIGMA retains a more stable label.

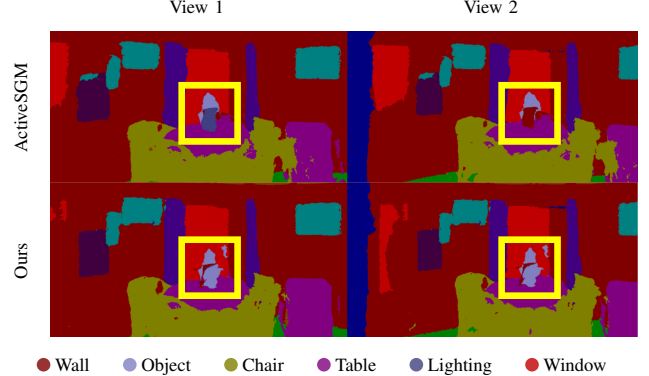


Fig. 3. Novel-view semantic rendering comparison in MP3D. The yellow boxes highlight a table-top vase whose label changes across views with ActiveSGM but remains stable with ActiveSIGMA.

D. Ablation Study

We conduct an ablation study on Replica to evaluate the contribution of the geometric, semantic uncertainty, and semantic inconsistency gains in the proposed NBV objective. All variants use the same mapping backend, semantic model, candidate generation procedure, exploration budget, and evaluation protocol.

As shown in Table II, removing either semantic gain degrades semantic mapping performance, while removing both results in the lowest mIoU. In particular, removing the semantic inconsistency gain reduces both mIoU and mAP, showing that cross-view inconsistency provides complementary guidance beyond semantic uncertainty. Removing the geometric gain also degrades overall performance, confirming the importance of sufficient geometric coverage. The complete objective achieves the best PSNR, SSIM, mIoU, and mAP, demonstrating the benefit of combining all proposed gains.

IV. CONCLUSIONS

We presented ActiveSIGMA, a semantic inconsistency-guided active mapping framework for 3D Gaussian Splatting. ActiveSIGMA maintains direction-conditioned Dirichlet evidence in a hybrid Gaussian-voxel map and converts cross-view semantic inconsistency into exploration heat for candidate generation and NBV scoring. Experiments on Replica and MP3D demonstrate improved semantic mapping performance and more consistent predictions across novel

viewpoints. These results demonstrate the effectiveness of cross-view semantic inconsistency as an exploration signal for active semantic mapping.

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